

Size, Atmosphere and Magnetic Field

Earth's size lets it hold an atmosphere; oxygen, the ozone layer and the magnetic field together shield and sustain life.

More than just distance

The right size

Earth is the **right size** so its **gravity** can hold on to an **atmosphere**.

- Too small → weak gravity, gases escape (Mars's atmosphere is ~100× thinner; Mercury has none).
- Too large → gravity so strong it could crush us.

The atmosphere and ozone

- **Oxygen** in the air lets almost all life breathe.
- Some oxygen forms **ozone** (a three-atom oxygen molecule), making the **ozone layer** that blocks harmful **ultraviolet (UV)** rays.

The magnetic field

The movement of **molten iron in Earth's core** makes Earth behave like a giant magnet. The **magnetic field** acts as a shield, deflecting harmful particles — **cosmic rays** (from space) and the **solar wind** (from the Sun) — protecting the atmosphere and life.

Together: right position + right size + atmosphere + ozone + magnetic field make Earth able to support and protect life.

Where students lose marks

Trap 1 — Gravity (from the right size) holds the atmosphere. A too-small planet loses its gases to space.

Trap 2 — Ozone blocks UV rays. Ozone is made of oxygen but is a different molecule (three oxygen atoms).

Trap 3 — The magnetic field shields from cosmic rays/solar wind, not from meteors. It comes from molten iron in the core.

Model answers — write them like this

Example A. Why does Earth hold an atmosphere but Mercury does not?

Step 1: Holding an atmosphere needs enough gravity.

Step 2: Earth's size gives enough gravity; small Mercury's is too weak, so its gases escaped.

∴ Earth's larger size and stronger gravity hold the atmosphere.

Example B. How does the ozone layer protect life?

Step 1: Ozone forms a layer high in the atmosphere.

Step 2: It absorbs harmful ultraviolet (UV) rays from the Sun.

∴ It shields living cells from damaging UV radiation.

[Ready to practise?](#) → [Open the worksheet](#)